مدينة زويل للعلوم والتكنولوچيا
Zewail City of Science and Technology

SPC 202 (Mechanics of Materials)

Bridge (phase 2)

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Introduction:

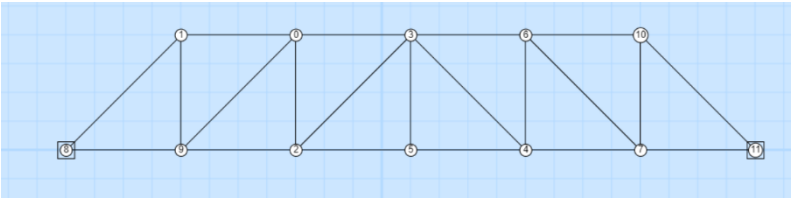
There are many kinds of bridges. When choosing an appropriate design for a bridge, one should examine the bridge's function, materials, topography of the location, and financial constraints. Any bridge that is built must have the least amount of deformation possible in order to handle large weights. The "truss bridge" is one of the many kinds of bridges. A single-span truss bridge is similar to a simply supported beam in that it distributes vertical stresses through bending. Bending affects the horizontal or vertical members, and, depending on their direction, either tension or compression in the vertical and diagonal members. Trusses are widely used and thought to be the best type because they use very little material to support rather big loads. It can withstand significant compression or tension. Furthermore, the materials readily accessible are suitable for the truss bridge.

Prototype Design & type of the Bridge:

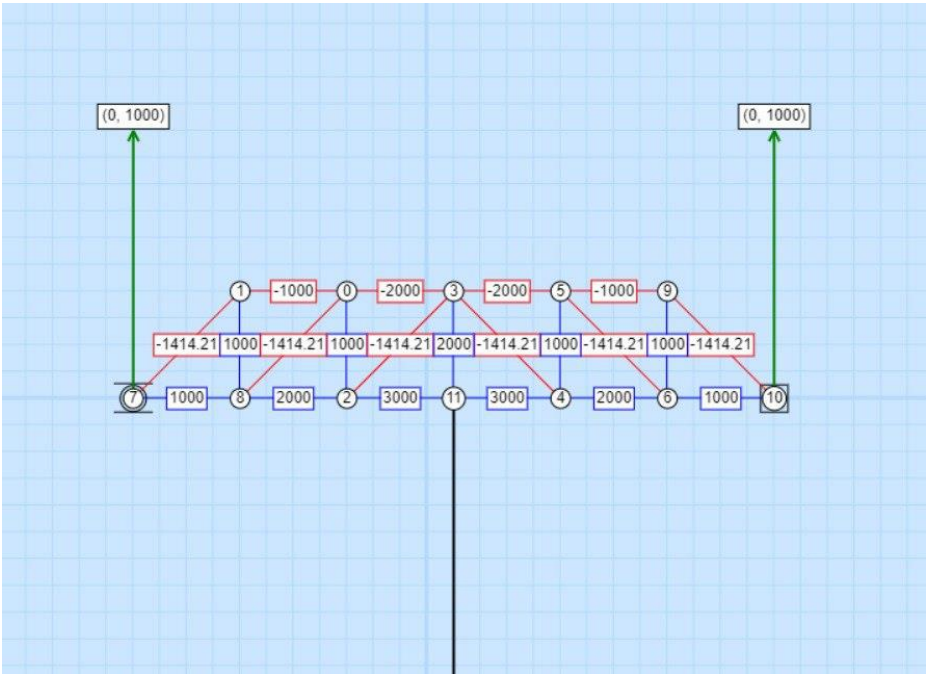
- Type of the bridge: In this project, we're going to create a Howe truss bridge using super glow, popsicle sticks, and strings.
  Advantages of Howe Truss:
  - The wood sticks' adaptability and ease of use: to demonstrate more using sticks allowed us to maintain the dimensions sufficiently before cutting on a wood slice.
  - The second crucial feature is its low weight, which we refer to as a weight-to-strength ratio that has been adequate since the Howe truss bridge aid in the distribution of its load across the entire bridge.
  - The capacity to adapt to conditions like extreme weather and in difficult areas, such as those above mountains.

  Disadvantages of Howe Truss:
  - The correct connection between the joints and bridge maintenance is so important.
  - The location it will be in because of its standard and spatial requirements it has a large and complicated structure.

- Design of the bridge:



- Analysis of the bridge:
  - This bridge would be constructed by 96 sticks.
  - Each member in the following diagram consists of two sticks glued together.
  - The following diagram shows only one side of the bridge. Hence, after dividing the shown force over the two glues together sticks, the forces would be divided over the two sides of the bridge.
  - The maximum load on a member in the proposed design is 323N.



Nodes

#	x [cm]	y [cm]
0	-3	4
1	-7	4
2	-3	0
3	1	4
4	5	0
5	5	4
6	9	0
7	-11	0
8	-7	0
9	9	4
10	13	0
11	1	0

Add new node

Members

#	Nodes	Length [cm]	Force [N]
0	0-1	4	-1000
1	0-2	4	1000
2	0-3	4	-2000
3	1-7	5.657	-1414.2136
4	1-8	4	1000
5	7-8	4	1000
6	2-8	4	2000
7	0-8	5.657	-1414.2136
8	2-3	5.657	-1414.2136
9	3-4	5.657	-1414.2136
10	3-5	4	-2000
11	5-6	5.657	-1414.2136
12	4-6	4	2000
13	4-5	4	1000
14	5-9	4	-1000
15	6-10	4	1000
16	9-10	5.657	-1414.2136
17	6-9	4	1000
18	3-11	4	2000
19	2-11	4	3000
20	4-11	4	3000

External Forces

Node	Fx [N]	Fy [N]
11	0	-2000
7*	0	1000
10*	0	1000

*Reaction forces at a support

Material Testing Results:

The material used in this project is the tongue depressors (wooden sticks). Since wood is an anisotropic material, the direction of the applied force determines the yield strength of the stick. In this bridge, only three forces were important to study. These forces are the stress and compression over the thickness-width area, and the buckling over the length-width area. These forces are examined as following:

1. Tension test: determining the tensile strength up till failure and breakage by using a tensile tester and a tensile force.
   Failure and breakage at 41 Kg which is at force = 402.21 N
   Tensile strength=23.66 MPa





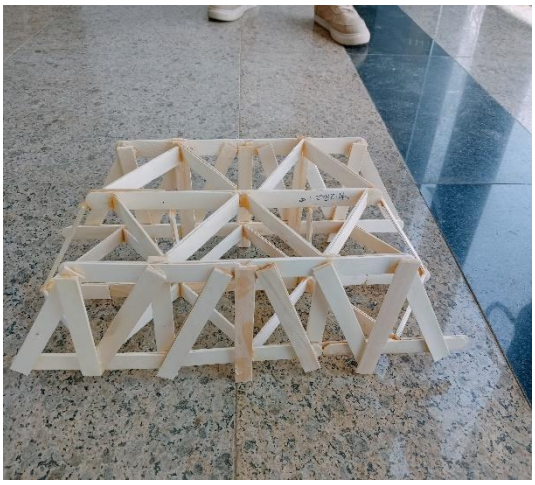
2. Compression test: As of the curvature of the sticks, making a compression test needs a specialized tools so that the applied force has no component that can cause buckling. Hence, this test was not made; however, as the stick is made of a solid, it is sensible that the compression yield of the stick would be equal or higher than tension yield.

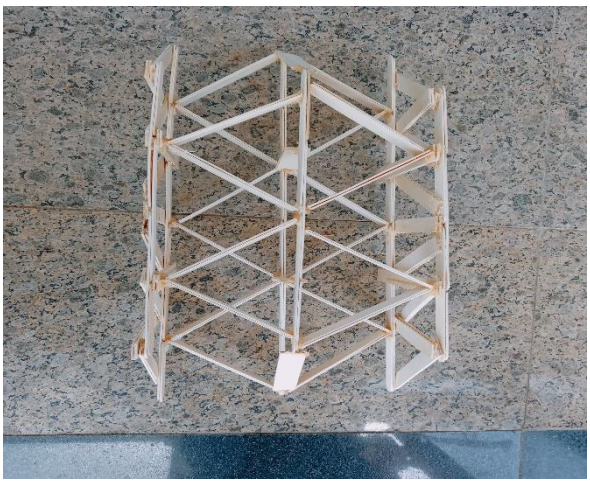
3. Buckling test: the buckling test was made by fixing the two ends of the stick all attaching weights on the middle of the length-width area of the stick. The measured buckling was 17.6N.



Final design of the Bridge:

We're going to create a Howe truss bridge





The Build:

- We use the glue and sticks as a material to make the bridge.
- We were keen to make the sticks double so that they can carry more weight.





The test of the maximum:

The maximum weight that the bridge was able to carry was 32.446 kg and the weight of the bridge itself was 240g, and this did not achieve the desired value due to several errors:

- The adhesive method itself was wrong using glue,
- There were also a lot of voids, and
- The weight of the bridge increased due to the use of more wood to no avail.
- Note that an amendment will be made to ensure that each member will carry a greater load, only to be doubled, not all of the members,
- And also, for the entire bridge to be one piece, without it. Additions



